Joint Supervised Classification and Reconstruction of Irregularly Sampled Satellite Image Times Series

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***Abstract*— Recent satellite missions have led to a huge amount of Earth observation data, most of them being freely available. In such a context, satellite image time series have been used to study land use and land cover information. However, optical time series, such as Sentinel-2 or Landsat ones, are provided with an irregular time sampling for different spatial locations, and images may contain clouds and shadows. Thus, preprocessing techniques are usually required to properly classify such data. The proposed approach is able to deal with irregular temporal sampling and missing data directly in the classification process. It is based on Gaussian processes and allows to perform jointly the classification of the pixel labels as well as the reconstruction of the pixel time series. The method complexity scales linearly with the number of pixels, making it amenable in large-scale scenarios. Experimental classification and reconstruction results show that the method does not compete yet with state-of-the-art classifiers but yields reconstructions that are robust with respect to the presence of undetected clouds or shadows and does not require any temporal preprocessing.**

***Index Terms*— Classification, Earth observation, Gaussian processes (GPs), irregular sampling, reconstruction, remote sensing, satellite image time series (SITS), Sentinel-2.**

NOMENCLATURE

A. Symbols and Notations

Symbol	Meaning
$\sim$	Distributed according to (a probability distribution).
$\mathcal{GP}$	Gaussian process: $f \sim \mathcal{GP}(m, K)$, the function f is distributed as a GP with mean function m and covariance operator K. The function is indexed in time by t.
$\mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	d-variate Gaussian (Normal) distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.
$f^d_{\boldsymbol{\mu},\boldsymbol{\Sigma}}$	Density associated with $\mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
$y\|x$	Random variable y conditionally on x.
$p(y\|x)$	Probability distribution of $y\|x$.
$\mathcal{S}$	Set of SITS or pixels.
n	Total number of SITS available in $\mathcal{S}$.
p	Number of wavelenghts of an SITS.
b	Band number, in $\{1, \ldots, p\}$.
$\mathcal{T}$	Temporal window where SITS are observed.
Y	$p-$dimensional process: $\mathcal{T} \to \mathbb{R}^p$.
$\mathbf{y}_{i,b}$	ith pixel from wavelength b.
T_i	Number of temporal acquisitions associated with pixel i.
$\mathcal{C}$	Number of classes.
c	Class value, in $\{1, \ldots, \mathcal{C}\}$.
z_i	Discrete random variable associated with pixel i and representing its class membership.
π_c	Prior probability of class c.
$\boldsymbol{\alpha}$, $\boldsymbol{\theta}$	Model parameters.
J	Number of basis functions, i.e., dimension of $\boldsymbol{\alpha}$.
$\|K\|$	Determinant of the matrix K.

I. INTRODUCTION

IN THE last decade, the successful launching of two satellites Sentinel-2 A and B offers a unique opportunity to record, analyze, and predict the evolution of the Earth's land surface. Sentinel-2 mission provides a high-resolution multispectral (13 spectral bands at 10, 20, or 60 m/pixel) acquisition with a five-day revisit cycle [1]. It is planned for long-term operational observations (more than 15 years), and optical Sentinel-2 satellite image time series (SITS) are available to users under a free and open data policy. Such a mission provides several terabytes of worldwide data per day [2].

Due to their spectral content and frequent update, SITS has found applications in various scientific fields: water management [3], [4], agricultural systems mapping [5]–[7], urban area analysis [8], [9], or ecological monitoring [10], [11]. However, such abundance of images raises new challenges in terms of large-scale multiresolution SITS processing [12]. Issues related to Big Earth Data were obviously explored in the

Manuscript received March 3, 2021; revised April 19, 2021; accepted April 23, 2021. Date of publication May 11, 2021; date of current version December 29, 2021. This work was supported in part by the French National Research Agency in the Framework of the Investissements d'Avenir Program under Grant ANR-15-IDEX-02 and in part by the Centre National d'Etudes Spatiales (CNES). *(Corresponding author: Alexandre Constantin.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2021.3076667 provided by the authors.

Digital Object Identifier 10.1109/TGRS.2021.3076667

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remote sensing community in order to calibrate and distribute the images seamlessly [13]–[17].

Image and signal processing issues were also investigated specifically for SITS properties. Spatial differences in terms of resolution were tackled using super-resolution strategies to resample all the spectral bands at the same finest spatial resolution (10 m) [18]–[20]. Fusion with other sensors, such as Landsat [21] or Sentinel-1 [22], was also considered to complement Sentinel-2 satellite images.

Classification of land cover or land use using machine learning techniques has emerged as a major application of SITS. Deep learning was intensively studied either using only one or few temporal acquisitions with convolutional neural networks (CNNs) [23], [24] or using the full temporal stack of acquisitions using recurrent neural networks (RNNs) [25], [26]. Combination of RNN applied on SITS with CNN applied on SPOT-6 image was also investigated for land cover mapping over Reunion Island [27]. Temporal convolution was investigated in [28]. Yet, due to the huge amount of data to be processed when the large geographical area or large temporal domain (e.g., one year of acquisition) is considered, lighter machine learning techniques, such as random forest (RF), were proposed as well, with very good results in terms of classification accuracy [29], [30]. Similarly, Derksen *et al.* [31] have shown that, when the number of spectrotemporal measurements per pixel is high, conventional handcrafted features and RF perform as well as CNN in terms of classification accuracy but with a drastically reduced processing time.

One major issue arising when dealing with large SITS, i.e. using several tracks or tiles,[1] is related to the irregular temporal sampling. Indeed, each Sentinel-2 track is acquired on different dates, and using images from different tracks results in pixels with different acquisition times. For instance, Fig. 1 shows the acquisition dates for three Sentinel-2 tiles located over three parts of France. Even though some dates are similar, the total number of acquisitions is different from one tile to another. Furthermore, clouds and shadows, occurring at random, are another cause of irregular temporal sampling. Dates with shadow/cloud are usually considered as missing values and are not considered to build the feature vector of the corresponding pixels. Hence, even pixels belonging to the same tile can be represented by feature vectors of different sizes [30].

Classical machine learning algorithms take as input a time series of finite and constant size. RF, support vector machine (SVM), or RNN use a vector representation of input features, i.e., a time series, or pixel, is a collection of features stacked into a p-dimensional vector and each pixel has the same number of features. Similarly, CNNs use a constant size patch representation of the image. Therefore, a preprocessing step, sometimes called missing information reconstruction, is usually applied on the SITS to recover pixels of the same size [32]. Among the various existing techniques, temporal filters, such as the Savitzky–Golay filter [33] or the best index slope extraction (BISE) [34] and parametric or nonparametric curve fitting [35]–[38], are popular due to their simplicity and efficiency. At a large scale (entire metropolitan France territory), linear temporal interpolation has shown to perform very well compared to spline interpolation [30]. Linear interpolation may also be combined with nonparametric smoothers [39].

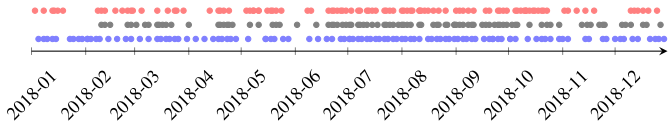


Fig. 1. Acquisition dates for tile T31TDN (red), T31TCJ (black), and T31TGK (blue) in 2018. The location of the tiles is given in Fig. 5. Large temporal gaps between two dates correspond to very cloudy acquisitions: images tagged as composed of more than 90% of cloudy pixels are not processed by the French data provider.

Methods taking into account both spatial and temporal information have provided interesting results (see nonlocal filters [40] or deep learning-based approaches [41]–[43]). However, due to their high computational cost, they are hardly applicable in national-/continental-scale settings [31]. Furthermore, they are also known as "black box" and hardly interpretable.

Recently, Gaussian processes (GPs) have gained attention in the remote sensing community [44], [45], GP mixed Bayesian and kernel methods to build statistical learning machines, for classification or regression problems. They have proved to be accurate and interpretable through their hyperparameters (mean and covariance functions) [46]–[49]. In this work, a GP model is introduced, relying on a linear projection of the mean function on a well-chosen basis, conditionally to the classes of interest. It is able to deal with irregularly sampled pixels in the learning and classification steps as well as reconstructing missing data. The contribution of this work is threefold.

1) The definition of a GP model that handles irregular temporal sampling (Section II), without resampling all the data on a common temporal grid.
2) Jointly to the classification, the proposed model allows for the reconstruction of pixels on any temporal grid, together with confidence intervals (Section III). Furthermore, the mean and covariance functions associated with each class are available for analysis.
3) The reconstruction used the GP model with optimal parameters (from the likelihood point of view) and considers the class membership. Rather than relying only on the reconstruction error, as with conventional missing data reconstruction techniques, the proposed method used the classification model likelihood to control the reconstruction and its subsequent smoothing level.

The remainder of this article is organized as follows. The proposed GP model and its estimation are described in Section II. The associated classification and reconstruction methods are derived in Section III. Section IV details the SITS datasets, while the experimental setup is presented in Section V. Experimental classification and reconstruction results are provided in Sections VI and VII, respectively. Finally, Section VIII concludes this article and opens discussion on future work.

[1]Sentinel-2 products are available as a collection of elementary tiles of size 100 × 100 km, see https://sentinel.esa.int/web/sentinel/missions/sentinel-2/data-products.

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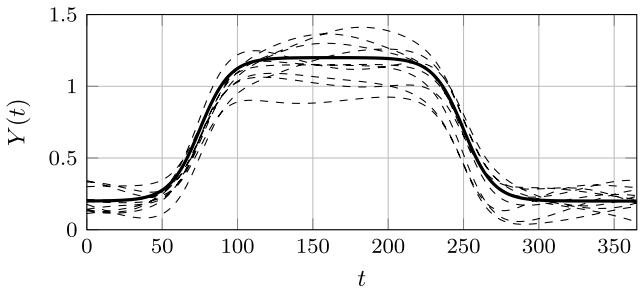


Fig. 2. Simulated univariate GPs with squared exponential covariance function (length scale parameter set to 50). The continuous black line is the mean function and the dashed lines are the corresponding ten realizations.

II. Irregularly Sampled GPs Model

In the following, $\mathcal{S} = \{(\mathbf{y}_i, z_i)\}_{i=1}^n$ denotes a set of n independent and identically distributed (i.i.d.) random multivariate and irregularly sampled pixels from an SITS, with their associated class labels (see the Nomenclature for an overview of the main notations. A pixel $\mathbf{y}$ is modeled by a random vector Y containing p random square integrable processes $\mathcal{T} \to \mathbb{R}^p$, where $\mathcal{T} = [t_{\min}, t_{\max}]$ is the time interval where the SITSs are observed

$$\begin{aligned} Y : \mathcal{T} &\to \mathbb{R}^p \\ t &\mapsto \big[Y_1(t), \ldots, Y_b(t), \ldots, Y_p(t)\big]^\top. \end{aligned}$$

This property is denoted by $Y \in L_2^p(\mathcal{T})$. The associated class z is modeled by a discrete random variable Z with possible values in $\{1, \ldots, \mathcal{C}\}$. In the context of this work, p is the number of wavelengths and/or spectral indices, $t_{\min} = 0$ and $t_{\max} = 365$ (in day-of-year unit).

A (univariate) GP f is a stochastic process such that any finite-dimensional marginal follows a multivariate Gaussian distribution [50]. It is specified by its mean function m and covariance function K

$$\begin{aligned} m(t_1) &= \mathbb{E}[f(t_1)] \\ K(t_1, t_2) &= \mathbb{E}[(f(t_1) - m(t_1))(f(t_2) - m(t_2))] \end{aligned}$$

with $t_1 \in \mathcal{T}$ and $t_2 \in \mathcal{T}$ two given times where the SITSs are observed, and we note $f \sim \mathcal{GP}(m, K)$. The covariance function K usually encodes *a priori* knowledge about the phenomena to be modeled. Fig. 2 shows several realizations of a GP with a squared exponential covariance function (see Section II-B2).

In the following, a multivariate GP model is proposed to cope with the multivariate nature of SITS and their irregular temporal sampling.

A. Mixture of Independent Multivariate GPs Model

Definition 1 Mixture of Independent Multivariate GPs: The proposed model, namely mixture of independent multivariate GP (MIMGP), relies on two main assumptions.

- *A1:* Each process Y_b, conditionally to $Z = c$, follows a GP: $Y_b | Z = c \sim \mathcal{GP}(m_{b,c}, K_{b,c})$.
- *A2:* Conditionally to $Z = c$, all components Y_b of Y are independent.

Here, $b \in \{1, \ldots, p\}$ is the spectral band and $c \in \{1, \ldots, \mathcal{C}\}$ is the class. The first property states that each band of a pixel from a given class is a realization of a univariate GP with class and band specific mean and covariance functions. The second property is introduced for computational reasons and is discussed in Section VI-B.

Let $\mathbf{y}_i$ be an observed pixel i at times $\{t_1^i, \ldots, t_{T_i}^i\}$. Its bth coordinate is represented by a vector of dimension T_i:

$$\mathbf{y}_{i,b} = \big[Y_b^i(t_1^i), \ldots, Y_b^i(t_{T_i}^i)\big]^\top.$$

By definition, conditionally to $Z = c$, this vector follows a T_i-variate Gaussian distribution:

$$\mathbf{y}_{i,b} | Z_i = c \sim \mathcal{N}_{T_i}\big(\boldsymbol{\mu}_{i,b,c}, \boldsymbol{\Sigma}_{b,c}^i\big) \tag{1}$$

where $\mathcal{N}_{T_i}$ is the Gaussian distribution on $\mathbb{R}^{T_i}$ with vector mean $\boldsymbol{\mu}_{i,b,c} = [m_{b,c}(t_1^i), \ldots, m_{b,c}(t_\ell^i), \ldots, m_{b,c}(t_{T_i}^i)]^\top$ and covariance matrix $(\boldsymbol{\Sigma}_{b,c}^i)_{\ell,\ell'} = K_{b,c}(t_\ell^i, t_{\ell'}^i)$.

Finally, in view of the independence assumption on the p wavelengths, $\mathbf{y}_i$ follows a product of p Gaussian densities:

$$\mathbf{y}_i | Z_i = c \sim \prod_{b=1}^{p} \mathcal{N}_{T_i}\big(\boldsymbol{\mu}_{i,b,c}, \boldsymbol{\Sigma}_{b,c}^i\big). \tag{2}$$

B. Mean and Covariance Functions

The unknown parameters of the model are the mean functions $m_{b,c}$ and the covariance operators $K_{b,c}$, for each band $b \in \{1, \ldots, p\}$ and each class $c \in \{1, \ldots, \mathcal{C}\}$. To deal with irregularly sampled observations, these parameters should be defined (and estimated) for any time t in the interval $\mathcal{T}$. To this end, projection techniques and parametric models are adopted, respectively, for the mean and the covariance functions. Parametric modeling of the latter ones is usual with GP (see [50, Chapter 4]). However, in our specific problem, using a projection method for the mean function allows to cope nicely with the irregular temporal sampling.

1) Mean Function: Let $\{\varphi_j\}_{j=1}^J$ be a subset of basis functions of $L_2(\mathcal{T})$. Then, $m_{b,c}$ can be written as

$$m_{b,c}(t) = \sum_{j=1}^{J} \alpha_{b,c,j} \varphi_j(t) + \varepsilon_{b,c}(t), \quad t \in \mathcal{T}$$

where $\alpha_{b,c,j}$ is the projection coefficient of $m_{b,c}$ on φ_j and $\varepsilon_{b,c}$ is an approximation error term.

Any functional basis can be used, e.g., Fourier, exponential, and splines (see [51, Ch. 5]). In case of the Fourier basis, $\alpha_{b,c,j}$ represents the amplitude of the corresponding frequency in the Fourier expansion. In the following, the basis is assumed to be fixed, while J, the number of basis functions, is a hyperparameter.

Introducing $\boldsymbol{\alpha}_{b,c} = [\alpha_{b,c,1}, \ldots, \alpha_{b,c,J}]^\top \in \mathbb{R}^J$ and $\mathbf{B}^i$ the $T_i \times J$ design matrix associated with pixel $\mathbf{y}_i$ defined by $(\mathbf{B}^i)_{\ell,j} = \varphi_j(t_\ell^i)$ with $\ell \in \{1, \ldots, T_i\}$, the mean vector in (1) can be written as

$$\boldsymbol{\mu}_{i,b,c} = \mathbf{B}^i \boldsymbol{\alpha}_{b,c}. \tag{3}$$

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2) Covariance Function: The covariance function is modeled using functions issued from the GP literature [50, Ch. 4]. A typical one would be a squared exponential covariance function with an additional colored noise [52] covariance function

$$K_{b,c}(t,s) = \gamma_{b,c}^2 \exp\left\{-\frac{(t-s)^2}{2h_{b,c}^2}\right\} + \sigma_{b,c}^2(t)\delta_{t,s}. \quad (4)$$

Introducing $\boldsymbol{\theta}_{b,c} = \{\gamma_{b,c}^2, h_{b,c}^2, \boldsymbol{\sigma}_{b,c}^2\}$, the covariance matrix in (1) is denoted in the following by:

$$\boldsymbol{\Sigma}_{b,c}^i = \boldsymbol{\Sigma}^i\left(\boldsymbol{\theta}_{b,c}\right). \quad (5)$$

Parameters $\boldsymbol{\alpha}_{b,c}$ and $\boldsymbol{\theta}_{b,c}$ are estimated by maximizing the marginal log likelihood, as explained in Section II-C.

C. Estimation

By plugging (3) and (5) in (2), it follows that:

$$\mathbf{y}_i | Z_i = c \sim \prod_{b=1}^{p} \mathcal{N}_{T_i}\left(\mathbf{B}^i \boldsymbol{\alpha}_{b,c}, \boldsymbol{\Sigma}^i\left(\boldsymbol{\theta}_{b,c}\right)\right). \quad (6)$$

Such an expression is a consequence of the use of nonzero mean GPs [50, Sec. 2.7]. The associated negative marginal log likelihood is given by

$$\ell(\boldsymbol{\alpha}, \boldsymbol{\theta}) = \sum_{b,c=1}^{p,C} \ell_{b,c}\left(\boldsymbol{\alpha}_{b,c}, \boldsymbol{\theta}_{b,c}\right)$$

with

$$\begin{aligned}\ell_{b,c}&\left(\boldsymbol{\alpha}_{b,c}, \boldsymbol{\theta}_{b,c}\right) \\ &= \sum_{i|Z_i=c} \Big[\log\left|\boldsymbol{\Sigma}^i\left(\boldsymbol{\theta}_{b,c}\right)\right| + \left(\mathbf{y}_{i,b} - \mathbf{B}^i\boldsymbol{\alpha}_{b,c}\right)^\top \\ &\qquad \times \boldsymbol{\Sigma}^i\left(\boldsymbol{\theta}_{b,c}\right)^{-1}\left(\mathbf{y}_{i,b} - \mathbf{B}^i\boldsymbol{\alpha}_{b,c}\right)\Big] + \kappa\end{aligned}$$

where κ is a constant independent of the parameters. Since independence is assumed between each spectral component, the optimization ends up with $p \times C$ independent optimization problems

$$\left(\hat{\boldsymbol{\alpha}}_{b,c}, \hat{\boldsymbol{\theta}}_{b,c}\right) = \arg\min_{\boldsymbol{\alpha}_{b,c}, \boldsymbol{\theta}_{b,c}} \ell_{b,c}\left(\boldsymbol{\alpha}_{b,c}, \boldsymbol{\theta}_{b,c}\right). \quad (7)$$

Marginal likelihood optimization is common practice in GP regression, while for binary classification, Laplace approximation is usually employed (and more computationally demanding) [50, Ch. 2 and 3]. Therefore, to the best of our knowledge, the use of marginal likelihood for multiclass classification is novel is this context.

Each subproblem (7) is solved by gradient descent as usually done in likelihood optimization [50, Ch. 5]. The algorithm is based on an alternate optimization with respect to $\boldsymbol{\alpha}$ and $\boldsymbol{\theta}$ (see Algorithm 1). At iteration (k), the update rule for $\boldsymbol{\alpha}$ is given by

$$\boldsymbol{\alpha}_{b,c}^{(k+1)} = \left[\sum_{i|Z_i=c} \mathbf{B}^{i\top} \boldsymbol{\Sigma}^i\left(\boldsymbol{\theta}_{b,c}^{(k)}\right)^{-1} \mathbf{B}^i\right]^{-1} \left[\sum_{i|Z_i=c} \mathbf{B}^{i\top} \boldsymbol{\Sigma}^i\left(\boldsymbol{\theta}_{b,c}^{(k)}\right)^{-1} \mathbf{y}_{i,b}\right]. \quad (8)$$

Algorithm 1 Estimation of Model Parameters

Input : $\mathcal{S}, \boldsymbol{\alpha}^0, \boldsymbol{\theta}^0$
Output: $\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\theta}}$
for $c = 1$ **to** C **do**
  for $b = 1$ **to** p **do**
    repeat
      Update $\boldsymbol{\alpha}_{b,c}$ using (8);
      Do a gradient step w.r.t. $\boldsymbol{\theta}_{b,c}$ using (9);
    until $\ell_{b,c}(\boldsymbol{\alpha}_{b,c}, \boldsymbol{\theta}_{b,c})$ *has converged*;

If the design matrix has no redundancies, it can be shown that the matrix in the left-hand side of (8) is indeed nonsingular, provided that the number of basis functions J is smaller or equal to the total number of unique observations in the training set. A proof is given in Section I (see the Supplementary Material).

For $\boldsymbol{\theta}$, there is no closed-form expression and a gradient step is required. The gradient is computed using the following partial derivative with respect to each coordinate θ_m of $\boldsymbol{\theta}$:

$$\frac{\partial}{\partial \theta_m} \ell_{b,c} = \sum_{i|Z_i=c} \operatorname{tr}\left(\left(\boldsymbol{\Sigma}^i\left(\boldsymbol{\theta}_{b,c}^{(k)}\right)^{-1} - \boldsymbol{\beta}_i \boldsymbol{\beta}_i^\top\right) \frac{\partial \boldsymbol{\Sigma}^i\left(\boldsymbol{\theta}_{b,c}^{(k)}\right)}{\partial \theta_m}\right) \quad (9)$$

with $\boldsymbol{\beta}_i = \boldsymbol{\Sigma}^i(\boldsymbol{\theta}_{b,c}^{(k)})^{-1}(\mathbf{y}_i - \mathbf{B}^i\boldsymbol{\alpha}_{b,c}^{(k+1)})$. Such an optimization procedure applied on simulated data from Fig. 2 (i.e., $C = 1$ and $p = 1$) leads to the estimation of the mean function shown in Fig. 3.

As a final remark, the $p \times C$ optimization problems can be solved in parallel since parameters are not shared between classes and spectral components.

D. Numerical Complexity

Usual GPs have complexity that scales in $\mathcal{O}(n^3)$, making them unsuitable for very-large-scale problems. In contrast, the proposed method has a reduced complexity that scales in $\mathcal{O}(n(T_M^3 + J^3))$, where T_M is the maximal length of observed time series ($T_M \geq T_i, \ \forall i \in 1, \ldots, n$). The first term comes from the inversion of $\boldsymbol{\Sigma}^i(\boldsymbol{\theta}_{b,c})$ and the second term comes from the update rule (8). In our application, T_M and J are much smaller than n, typically by several orders of magnitude. Furthermore, MIMGP allows for the classification of any irregularly new time series as well as the reconstruction of the observed time series using reconstruction techniques, as described in the following.

III. Classification and Reconstruction of Missing Values

Once the parameters are estimated, it is possible to classify a new pixel without any temporal resampling, as for training. In addition, MIMGP is also able to reconstruct missing values on any temporal scheme.

In the following, $\mathbf{y}_j$ denotes a new SITS observed at T_j times denoted by $\{t_1, \ldots, t_{T_j}\}$, which may not have been observed in the training set.

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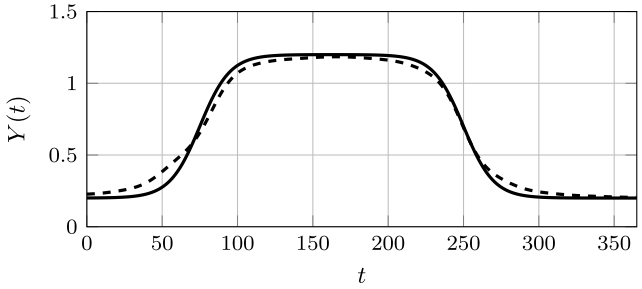


Fig. 3. Continuous line is the mean function associated with the GP shown in Fig. 2 and the dashed line is the estimated one using Algorithm 1 and $J = 20$ Gaussian functions (see Section V-A for details on bases).

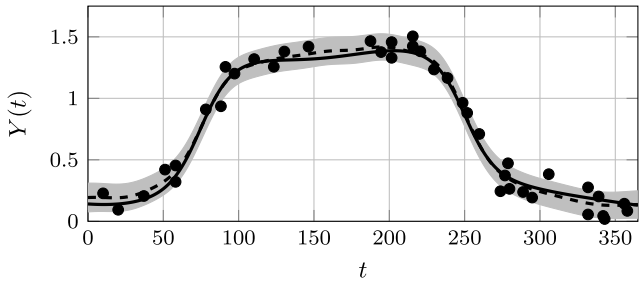


Fig. 4. Time-series reconstruction. The black continuous line is one (continuous) realization of the GP from Fig. 2. The black dots are the observed noisy acquisitions at several (discrete) times. The dashed line is the reconstructed time series. The gray region displays a pointwise confidence region for the reconstruction: $\hat{Y}_b^j(t^*) \pm \sqrt{\mathbb{V}}[\hat{Y}_b^j(t^*)]$.

A. Classification of a New Time Series

The *a posteriori* probability $\mathbb{P}(Z_j = c|\mathbf{y}_j)$ to belong to a class c given $\mathbf{y}_j$ is computed using the product of Gaussian densities given in (6) and Bayes' rule

$$\mathbb{P}\big(Z_j = c\big|\mathbf{y}_j\big) \propto \pi_c \mathbb{P}\big(\mathbf{y}_j\big|Z_j = c\big) \propto \pi_c \prod_{b=1}^{p} f_{T_j}\big(\mathbf{y}_{j,b}; \mathbf{B}^j \boldsymbol{\alpha}_{b,c}, \boldsymbol{\Sigma}\big(\boldsymbol{\theta}_{b,c}\big)\big) \quad (10)$$

where $f_d(\cdot; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the d-variate Gaussian density with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ and π_c is the *prior* probability $\mathbb{P}(Z = c)$ of the class $c \in \{1, \ldots, \mathcal{C}\}$.

In practice, π_c is estimated by its empirical counterpart $\hat{\pi}_c = n_c/n$, where n_c is the number of pixels assigned to class c in the set $\mathcal{S}$. Parameters $\hat{\boldsymbol{\alpha}}_{b,c}$ and $\hat{\boldsymbol{\theta}}_{b,c}$ are estimated due to Algorithm 1. The new time series $\mathbf{y}_j$ is then assigned to the class of maximum posterior probability (MAP rule)

$$\hat{z}_j = \max_c \mathbb{P}\big(Z_j = c\big|\mathbf{y}_j\big).$$

B. Time-Series Reconstruction

Time-series reconstruction is achieved using conditional expectations of Gaussian distributions. Two cases are considered. Either the class membership of the considered pixel is known or the class membership is estimated using the posterior probability.

1) Reconstruction When Class Membership Is Known: Let us write $Y_b^j(t^*)$ the unobserved value at wavelength b and time t^*. The following reconstruction rule, based on the conditional expectation, is considered:

$$\hat{Y}_{b,c}^j(t^*) := \mathbb{E}\Big[Y_b^j(t^*)\big|\mathbf{y}_j, Z_j = c\Big].$$

Using the properties of Gaussian distributions [53, p.63] and replacing the unknown quantities by their estimated counterparts yield (see Appendix A for a proof)

$$\hat{Y}_{b,c}^j(t^*) = \mathbf{b}^*\hat{\boldsymbol{\alpha}}_{b,c} + \mathbf{k}\Big(t^*, t_{1:T_j}^j\big|\hat{\boldsymbol{\theta}}_{b,c}\Big)^\top \boldsymbol{\Sigma}^j\big(\hat{\boldsymbol{\theta}}_{b,c}\big)^{-1}\big(\mathbf{y}_{j,b} - \mathbf{B}^j\hat{\boldsymbol{\alpha}}_{b,c}\big) \quad (11)$$

and

$$\mathbb{V}\Big[\hat{Y}_{b,c}^j(t^*)\Big] = K\big(t^*, t^*\big|\hat{\boldsymbol{\theta}}_{b,c}\big) - \mathbf{k}\Big(t^*, t_{1:T_j}^j\big|\hat{\boldsymbol{\theta}}_{b,c}\Big)^\top \times \boldsymbol{\Sigma}^j\big(\hat{\boldsymbol{\theta}}_{b,c}\big)^{-1}\mathbf{k}\Big(t^*, t_{1:T_j}^j\big|\hat{\boldsymbol{\theta}}_{b,c}\Big) \quad (12)$$

where $\mathbf{k}(t^*, t_{1:T_j}|\hat{\boldsymbol{\theta}}_{b,c}) = [K(t^*, t_1|\hat{\boldsymbol{\theta}}_{b,c}), \ldots, K(t^*, t_{T_j}|\hat{\boldsymbol{\theta}}_{b,c})]^\top$, $\mathbf{b}^* = [\varphi_1(t^*), \ldots, \varphi_J(t^*)]$ contains the evaluation of the basis functions at time t^*, and $\mathbf{B}^j$ is the $T_j \times J$ design matrix associated with $\mathbf{y}_j$.

Equations (11) and (12) provide, respectively, the reconstructed value and the variance of the reconstruction. They are given here for a single time t^*, but similar formulas can be derived for multiple times $t_1^*, \ldots, t_{T^*}^*$. Interestingly, (11) shows that the reconstruction is given by $\mathbf{b}^*\hat{\boldsymbol{\alpha}}_{b,c}$, the estimated mean of the GP, corrected by a value proportional to the error made at the acquisition time of pixel j. A similar remark holds for the variance. It can be interpreted as the estimated variance of the GP corrected by the variance of the process observed at the acquisition times of pixel j. Figure 4 shows a reconstructed timeseries using (11) and (12).

2) Reconstruction When Class Membership Is Unknown: The reconstruction of $\hat{Y}_b^j(t^*)$ is done as the average of the previous reconstructions $\hat{Y}_{b,c}^j(t^*)$ in each class [see (11)] weighted by the posterior probabilities estimated with (10)

$$\hat{Y}_b^j\big(t^*\big) = \sum_{c=1}^{\mathcal{C}} \mathbb{P}\big(Z_j = c\big|\mathbf{y}_j\big)\hat{Y}_{b,c}^j\big(t^*\big). \quad (13)$$

A similar formula holds for the variance with an additional between-class variance term

$$\mathbb{V}\Big[\hat{Y}_b^j\big(t^*\big)\Big] = \sum_{c=1}^{\mathcal{C}} \mathbb{P}\big(Z_j = c\big|y_j\big)\mathbb{V}\Big[\hat{Y}_{b,c}^j\big(t^*\big)\Big] + \sum_{c=1}^{\mathcal{C}} \mathbb{P}\big(Z_j = c\big|y_j\big)\Big[\hat{Y}_{b,c}^j\big(t^*\big)^2 - \hat{Y}_b^j\big(t^*\big)^2\Big]$$

where $\hat{Y}_{b,c}^j(t^*)$ and $\hat{Y}_b^j(t^*)$ are the reconstructions at t^*, respectively, when the class is known (11) and when the class is unknown (13), see Appendix A for a proof.

IV. Sentinel-2 SITS Datasets

Three Sentinel-2 tiles of level 2A over the French metropolitan territory were downloaded from the Theia Land Data Center.[2] All available acquisitions between January 2018 and December 2018 for the two orbits of satellites Sentinel-2A

[2]http://www.theia-land.fr/en/presentation/products

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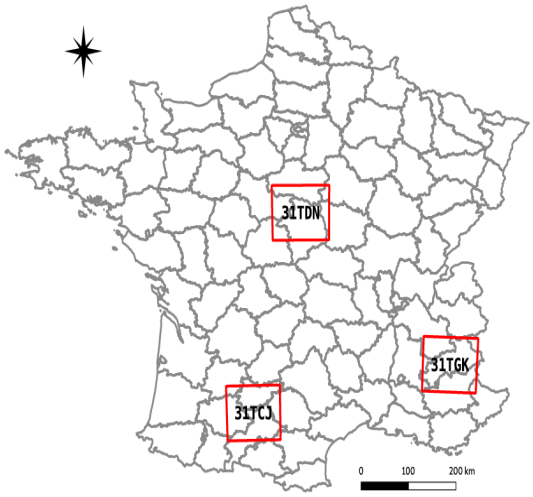


Fig. 5. Location of the Sentinel-2 tiles. The label inside the square is the tile name.

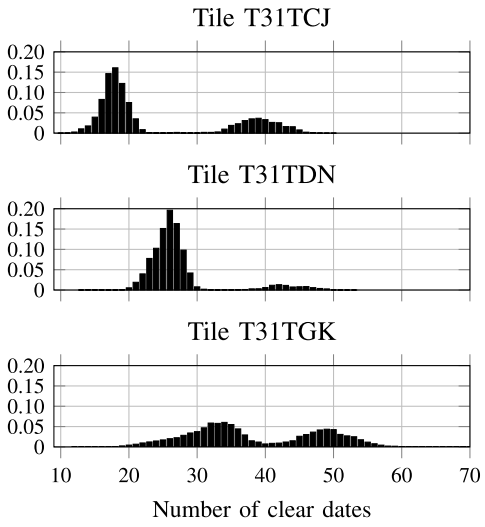


Fig. 6. Proportion of pixels as functions of the number of clear dates.

and 2B were used. Fig. 5 shows the location of the three tiles. They correspond to different climatic regions [30], with varying meteorological and topographical conditions.

Surface reflectance time series were produced using the Multisensor atmospheric correction and cloud screening-ATCOR Joint Algorithm (MAJA) processing chain developed by the CNES-CESBIO and DLR [17]. It involves orthorectification, atmospheric correction, clouds, and shadows detection. Spectral bands at 10 and 20 m/pixel were used, for a total of ten spectral bands. Bands at 20 m/pixel were spatially upsampled to 10 m/pixel using the Orfeo ToolBox[3] [54]. Fig. 6 shows the distribution of the number of clear dates (dates not tagged clouds, shadows, or no data in the raw time series): It appears that the number of clear acquisitions per pixel significantly varies depending on the tile.

Furthermore, the data from each tile were resampled and gap-filled (missing information due to clouds or shadows were reconstructed using linear interpolation) onto the same set of dates (every five days, starting from January 1, 2018, and ending December 27, 2018) as in [30]. Fig. 11 shows the raw data and the resampled data for one random pixel from each tile in the same class. Hence, for each pixel location, three temporal informations are available after the preprocessing.

1) The raw multispectral time series with irregular acquisition dates.
2) The mask time series, indicating for each acquisition date the presence/absence of clouds/shadows. In our model, the presence of clouds/shadows implies that the corresponding raw spectral values are considered as missing values.
3) The resampled multispectral time series with regular acquisition dates.

TABLE I
LAND COVER CLASSES AND NUMBER OF EXTRACTED SITS IN EACH TILE

Class	T31TCJ	T31TDN	T31TGK
Artificial areas			
Continuous urban fabric	10,000	8,292	959
Discontinuous urban fabric	10,000	10,000	10,000
Industrial or commercial units	10,000	10,000	10,000
Road surfaces	10,000	9,906	3,664
Agricultural areas			
Winter crops	30,000	30,000	15,975
Summer crops	40,000	40,000	24,912
Forest and semi-natural areas			
Meadow	10,000	10,000	10,000
Orchards	10,000	2,775	10,000
Vines	10,000	8,719	153
Broad-leaved forest	10,000	10,000	10,000
Coniferous forest	9,957	10,000	10,000
Natural grasslands	9,939	3,022	10,000
Woody moorlands	9,972	10,000	10,000
Open spaces with little or no vegetation			
Bare rock	0	0	10,000
Beaches, dunes and sand plains	0	5,355	10,000
Glaciers and perpetual snow	0	0	10,000
Water bodies	10,000	10,000	10,000
Total	189,868	178,069	165,663

The reference data arise from the work of Inglada *et al.* [30]. They were extracted from freely available data sources. Seventeen land cover classes were defined, ranging from artificial areas to vegetation and water bodies. Table I shows the exhaustive list of classes.

The reference data are provided as a set of spatial polygons overlapping the three tiles. Fig. 7 shows an extract of these polygons. The training and validation sets were constructed by stratifying pixels according to the polygons membership information. Pixels from one polygon fully belong to either the training or the validation set. Depending on the number of available referenced pixels per class, 10 000 (or less) pixels were extracted for the training and validation set, except for winter and summer crops, for which 30 000 and 40 000 pixels were extracted, respectively. Ten independent train/validation sets were generated for statistical validation. Table I shows the average number of pixels per class and per tile.[4]

[3]Using "SuperImpose" application, see https://www.orfeo-toolbox.org/CookBook/Applications/app_Superimpose.html

[4]Since a stratification with respect to polygons is done and polygons size can vary in terms of area, the number of pixels may vary slightly from one experiment to another. The model was trained on a total average of 178 000 pixels and was evaluated on 178 000 pixels.

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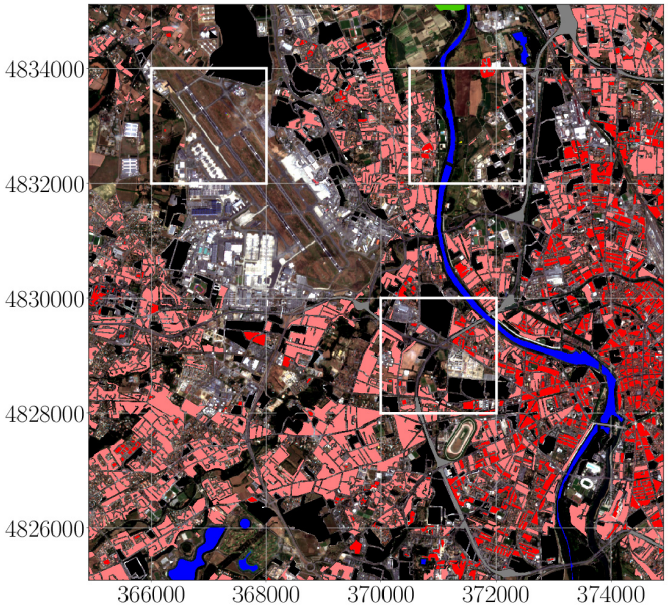


Fig. 7. Reference data extraction where each color represents an extract of land cover classes from Table I. The color code is described in Fig. 9. The axes correspond to the geographical coordinates. The white continuous squares correspond to zoomed areas of the classification thematic maps provided in Section VI. The background image is an extract from Sentinel-2 optical images.

V. Experimental Setup

The model parameters are $\boldsymbol{\alpha}_{b,c}$ and $\boldsymbol{\theta}_{b,c}$ for each class c and each wavelength b. They are estimated using the training set of pixels. MIMGP has also some hyperparameters that are set before the learning step: the design matrix $\mathbf{B}$ with the associated number of basis functions J and the family of parametric kernels for the covariance operator. These settings are common to each class and each wavelength.

A. Functional Bases

Four functional bases have been investigated. Two of them are local bases (nonzero only on subsets of $[0, \mathcal{T}]$), the Gaussian and BSplines bases, while the remaining ones are global bases, the well-known Fourier and Polynomial bases. A user-defined hyperparameter, denoted by J^*, is used to select the number of basis functions, as explained in the next paragraph. The choice of a basis setting corresponds to an assumption on the temporal behavior of the time series. For instance, a Fourier basis can represent a periodic signal in the time domain.

1) *Fourier:* The number of basis functions is $J = 2J^* + 1$ and

$$m(t) = \alpha_0 + \sum_{j=1}^{J^*} \left[\alpha_j \cos\left(2\pi j \frac{t}{\mathcal{T}}\right) + \alpha_{j+J^*} \sin\left(2\pi (j+J) \frac{t}{\mathcal{T}}\right) \right].$$

2) *Polynomial:* $m(t) = \alpha_0 + \sum_{j=1}^{J^*} \alpha_j t^j$, with $J = J^* + 1$.

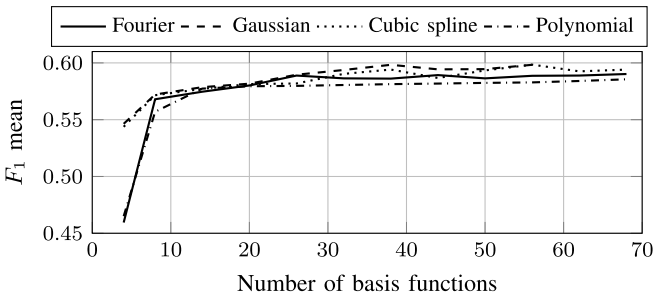


Fig. 8. Averaged "mean F_1 scores" of the MIMGP model with the four bases as a function of the basis dimension. The results are reported for the tile T31TCJ.

3) *Gaussian:*

$$m(t) = \sum_{j=1}^{J^*} \alpha_j \exp\left(-\frac{(t-t_j)^2}{d_j^2}\right)$$

with $J = J^*$. $t_j, j \in \{1, \ldots, J^*\}$ can be equidistant in $\mathcal{T}$ or chosen as quantiles of the distribution of the clean dates. The hyperparameter d_j is set such that $d_j^2 = 8|t_{j+1} - t_j|$ to ensure a sufficient overlap between two consecutive local exponential functions.

4) *Cubic Splines:* $m(t) = \sum_{j=0}^{J^*} \alpha_j S_j(t)$, with $J = J^* + 1$. S_j is the jth bicubic spline on $[0, \mathcal{T}]$. The knots t_j are chosen as in the Gaussian case.

The invertibility condition associated with (8) implies that J should be smaller than 92, 100, and 103 for tiles T31TCJ, T31TDN, and T31TGK, respectively.

B. Covariance Function

Any convex combination of positive semidefinite kernels is a valid covariance function [50, Ch. 4]. In this work, the squared exponential covariance function added with a colored noise covariance function, as given in (4), is used.

The parameters $\boldsymbol{\theta}_{b,c} = \{\gamma_{b,c}^2, h_{b,c}^2, \boldsymbol{\sigma}_{b,c}^2\}$, $\forall \{b, c\} \in [1, p] \times [1, C]$, are learned as described in Algorithm 1. The estimated covariance parameters provide some insights about the observed processes. In particular, the length scale $h_{b,c}$ is related to the temporal behavior of the reflectance. For a given band b and conditionally to class c, the longer $h_{b,c}$ grows, the more correlated two distant dates are.

VI. Supervised Classification

The classification accuracy is assessed by the F_1 score, computed as the harmonic mean of the precision and recall for each class. We report the "mean F_1 score," which is the mean of F_1 scores computed on all the classes of the dataset.

First, the influence of the basis functions on the classification score is investigated. Then, a comparison with other classifiers is reported and discussed. Convergence and model parameter analyses are provided in Section II (see the Supplementary Material).

A. Influence of the Basis Functions

The influence of the basis functions and its dimension is investigated. Fig. 8 shows the averaged "mean F_1 scores"

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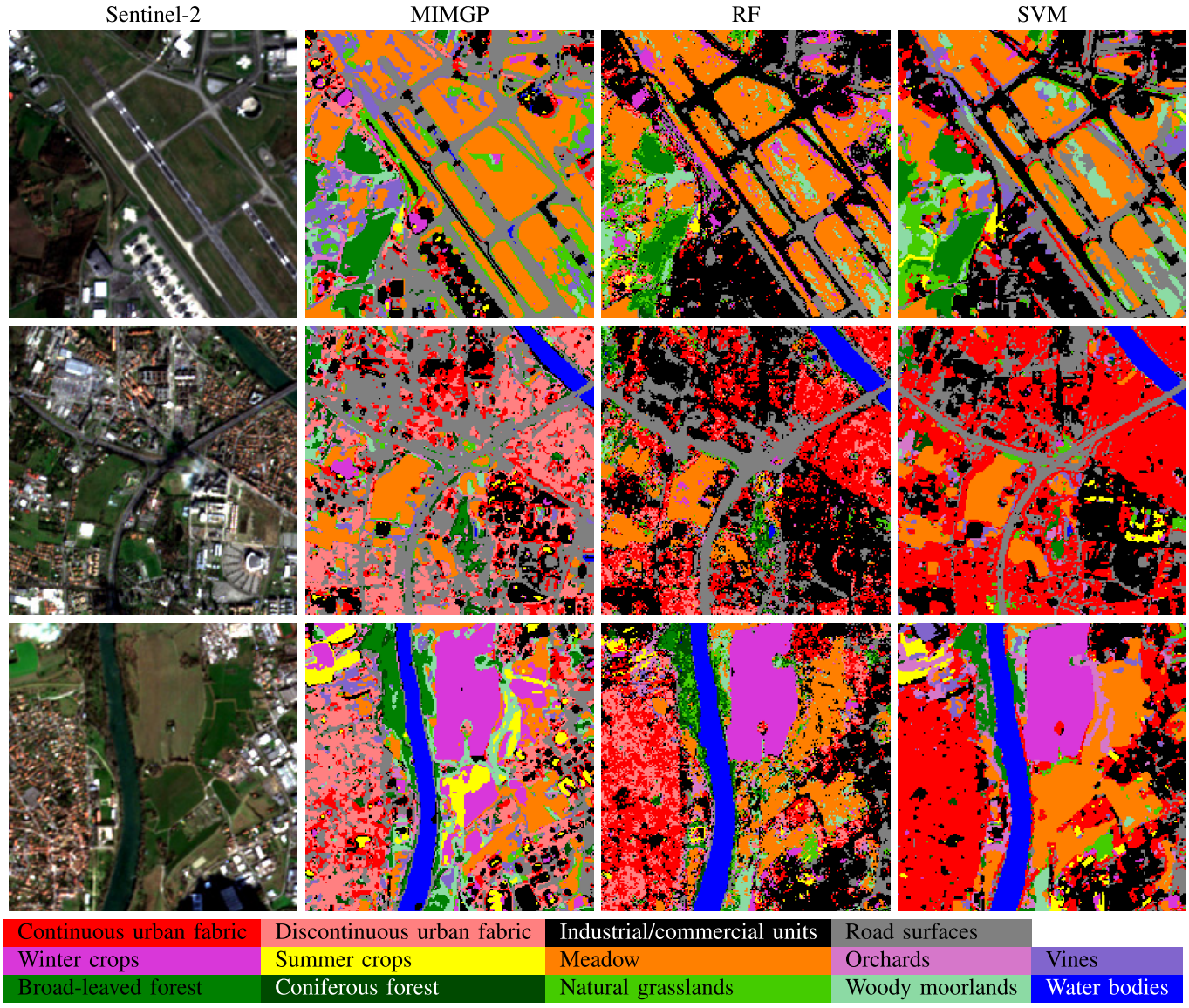


Fig. 9. Thematic maps for three sites from tile T31TCJ. These sites are located as shown in Fig. 7.

obtained on the ten independent train/validation datasets for the tile T31TCJ. From the figure, it appears that the dimension of the basis should be larger than 9 for this tile to allow for a reasonable classification score, independently of the basis itself. Furthermore, MIMGP is robust with respect to basis selection since similar accuracy is obtained for all bases.

However, numerical instability has been observed for a very large number of basis functions, in particular for the exponential basis. For instance, when the number of functions is greater than 50, consecutive local exponential bases overlap too much and the invertibility conditions are violated (see Section I in the Supplementary Material). In practice, the Fourier basis is the most stable and was used to compare our method with other classifiers.

B. Comparison With Other Classifiers

The performances of MIMGP are compared with three other methods: quadratic discriminant analysis (QDA) that involves a similar Gaussian assumption on regularly sampled data, a linear SVM fitted with a stochastic gradient descent (SGD) [55], and RFs [56] that have shown state-of-the-art results in large-scale pixelwise classification of SITS [31]. QDA, SVM, and RF methods take the resampled multispectral time series as input. The ten spectral bands are stacked together to let these three methods account for dependence between spectral bands.

RF is used with 100 trees and a maximum depth of tree set to 25. The MIMGP model is implemented with the Fourier basis and a Radial Basis Function (RBF) kernel with an additive colored noise, which represents a total of 32 parameters for $\boldsymbol{\theta}$.

Table II summarizes the results for the four methods.[5] We can see that some significant improvements are achieved compared with the standard QDA classifier, even though

[5]Per class F_1 scores are provided in the Supplementary Material.

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TABLE II

AVERAGED "MEAN F_1 SCORE" (MEAN(%) $\pm$ STANDARD DEVIATION) COMPUTED ON THREE TILES. MIMGP MODEL IS PARAMETERIZED BY A FOURIER BASIS WITH 19 PARAMETERS FOR T31TCJ AND 41 FOR T31TDN AND T31TGK TILES

	QDA	SVM	RF	MIMGP
T31TCJ	36.2 ± 2.5	70.1 ± 2.1	72.2 ± 1.6	57.6 ± 2.0
T31TDN	30.5 ± 2.8	74.8 ± 1.9	77.6 ± 1.5	65.1 ± 1.0
T31TGK	38.9 ± 2.1	62.1 ± 1.6	63.9 ± 1.6	45.5 ± 2.7

MIMGP imposes independence between wavelengths. The QDA classifier suffers from numerical instability due to the high number of spectrotemporal features. However, MIMGP is not as accurate as the RF and the SVM classifiers applied on the reconstructed time series.

Fig. 9 shows three extracts of the classification map for tile T31TCJ.[6] Visually, significant differences can be observed both in terms of homogeneity and in terms of pixelwise prediction. For instance, for the airport area, SVM predicts wrongly a lot of natural grasslands, while MIMGP and RF predict correctly meadow for vegetation between runways. Overall, MIMGP and SVM seem to predict more homogeneous maps than RF. This can be clearly seen in the second row, where the city center classification maps obtained with RF contain salt and pepper noise.

The results provided in this section show that MIMGP is not at the level of state-of-the-art methods, such as RF or SVM. One critical point is the spectral independence assumption that is not true in practice and used here for computational purposes. Removing this constraint would help to improve the classification accuracy.

Yet, the computation of MIMGP does not require any temporal resampling processes and directly handles the raw time series in the training and prediction steps. In comparison with standard GPs for classification, its complexity is linear with respect to the number of samples and cubic with respect to the number of temporal acquisitions, thus allowing large-scale processing. Furthermore, it can also be used to reconstruct pixel time series, as discussed in Section VII.

VII. TIME-SERIES RECONSTRUCTION

Following Section III-B, MIMGP is able to reconstruct missing values conditionally to a known, or predicted, class. Class mean functions and covariance operators are also obtained after the training step [for clarity, mean functions and covariance operators are provided in Section III (see the Supplementary Material)]. In the following, the quality of the reconstruction is compared with a state-of-the-art method: the Whittaker smoother [37], [39].

To assess the quality of the reconstruction, the validation set was used to ensure that reconstructed pixels were not seen during the training phase. We randomly remove one clear date from each pixel and reconstruct it with the trained MIMGP model and the Whittaker smoother for which the regularization parameter is set using the V-curve technique for each individual pixel [39], [57]. The quality of the reconstruction

[6]Full classification map and extracts for the three tiles are provided in the Supplementary Material.

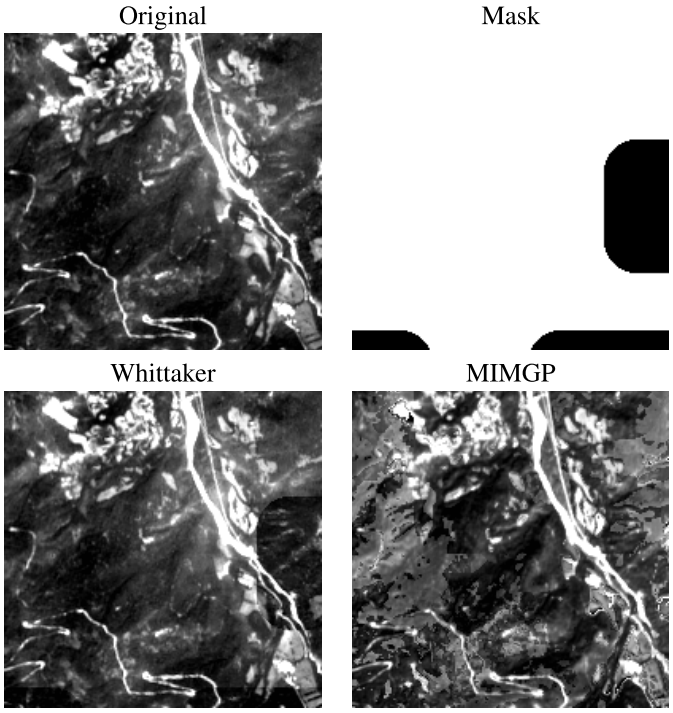


Fig. 10. Image reconstruction from a site in tile T31TGK. This site is located as shown in Fig. 5 (see the Supplementary Material). (Top Left) Original spectral band 4 (red) on August 6, 2018. (Top Right) mask: white color for a clear pixel and black color for pixel with clouds/saturation as detected by the preprocessing chain. (Bottom Left and Right) Reconstruction with the Whittaker smoother and the MIMGP, respectively.

is estimated using the normalized mean absolute error (nMAE) computed as

$$\text{nMAE} = \frac{\sum_{i=1}^{n_v} \left| Y_b^i(t^*) - \hat{Y}_b^i(t^*) \right|}{\sum_{i=1}^{n_v} \left| Y_b^i(t^*) - \overline{Y_b} \right|}$$

where $\overline{Y_b} = (1/n_v) \sum_{i=1}^{n_v} Y_b^i(t^*)$, with n_v the number of validation pixels. Table III summarizes the results obtained on the three tiles and for the ten independent datasets. The scatter plots associated with nMAE results are provided in the Supplementary Material.

From the table, MIGMP outperforms the Whittaker smoother in terms of nMAE for each band for tiles T31TDN and T31TGK. For tile T31TCJ, Whittaker is better for five bands. It is important to note that MIMGP provides better nMAE and does not need any additional fitting, apart the training step, in contrast to the Whittaker smoother that requires to be fitted for each pixel, as well its regularization parameter.

Fig. 10 shows a part of the tile T31TGK reconstructed for band 4 (red) on August 6, 2018. For this simulation, all the clear dates are used to reconstruct the time series for both methods. Interestingly, for the Whittaker smoother, the clouds mask is visible, indicating a noncontinuous reconstruction of the dynamic. This artifact is not visible for the proposed method.

Fig. 11 shows the red wavelength (R) for two pixels of the same class (summer crops). They were selected to enlighten the robustness of the proposed method to an inaccurate cloud mask file. The first pixel has an undetected cloud for the third

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TABLE III

AVERAGED NORMALIZED MAE (MEAN (%) ± STANDARD DEVIATION) FOR EACH WAVELENGTH FOR THE DIFFERENT TILES WITH THE A FOURIER BASIS OF DIMENSION $J = 19$

S2 Band	Method	T31TCJ	T31TDN	T31TGK
Band 2	Whittaker	47.1 ± 0.5	123 ± 1.5	40.1 ± 1.2
	MIMGP	$\mathbf{46.9} \pm 2.4$	$\mathbf{46.5} \pm 1.3$	$\mathbf{36.3} \pm 1.4$
Band 3	Whittaker	45.2 ± 0.5	114 ± 1.2	42.3 ± 1.3
	MIMGP	$\mathbf{43.9} \pm 3.7$	$\mathbf{44.0} \pm 1.6$	$\mathbf{39.6} \pm 1.4$
Band 4	Whittaker	$\mathbf{37.7} \pm 0.4$	94.2 ± 1.0	43.5 ± 1.3
	MIMGP	42.7 ± 4.4	$\mathbf{45.2} \pm 2.2$	$\mathbf{41.3} \pm 1.5$
Band 5	Whittaker	$\mathbf{40.0} \pm 0.5$	99.9 ± 1.0	44.7 ± 1.3
	MIMGP	45.8 ± 0.4	$\mathbf{40.9} \pm 2.0$	$\mathbf{41.3} \pm 2.2$
Band 6	Whittaker	$\mathbf{32.6} \pm 0.4$	83.2 ± 1.0	51.1 ± 1.3
	MIMGP	38.6 ± 1.8	$\mathbf{46.4} \pm 0.9$	$\mathbf{46.9} \pm 2.6$
Band 7	Whittaker	30.0 ± 0.3	73.5 ± 0.9	51.1 ± 1.2
	MIMGP	$\mathbf{24.7} \pm 2.7$	$\mathbf{45.1} \pm 0.9$	$\mathbf{41.8} \pm 1.1$
Band 8	Whittaker	$\mathbf{32.6} \pm 0.4$	71.8 ± 0.9	52.9 ± 1.2
	MIMGP	33.1 ± 1.9	$\mathbf{44.6} \pm 2.3$	$\mathbf{42.9} \pm 1.1$
Band 8A	Whittaker	$\mathbf{29.4} \pm 0.3$	69.3 ± 0.9	51.9 ± 1.2
	MIMGP	33.2 ± 3.4	$\mathbf{43.8} \pm 1.5$	$\mathbf{42.6} \pm 1.2$
Band 11	Whittaker	$\mathbf{36.1} \pm 0.5$	53.1 ± 0.7	46.9 ± 0.6
	MIMGP	37.0 ± 2.0	$\mathbf{38.3} \pm 1.7$	$\mathbf{43.0} \pm 1.3$
Band 12	Whittaker	$\mathbf{34.1} \pm 0.4$	46.0 ± 0.6	51.3 ± 0.7
	MIMGP	39.5 ± 2.6	$\mathbf{38.7} \pm 3.3$	$\mathbf{46.6} \pm 2.6$

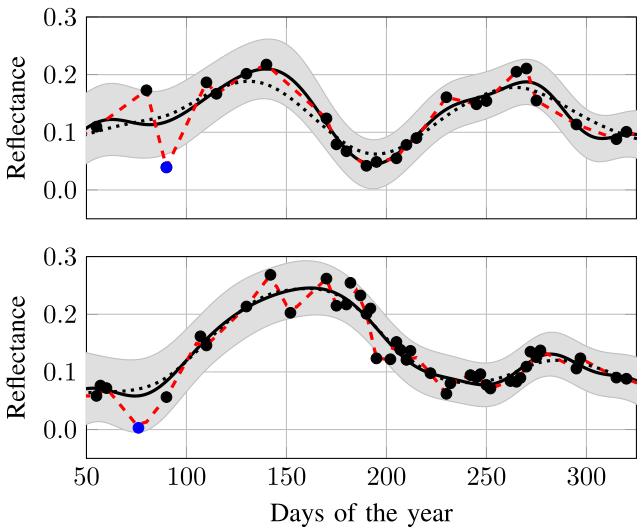


Fig. 11. Time-series reconstruction. (Top and Bottom) Red reflectance for two different pixels. Dot points are the original values from the irregularly sampled time series. Blue points are nondetected clouds and saturation by the satellite data preprocessing chain, for the top and bottom figures, respectively. The black continuous line is the conditional reconstruction of the signal and the gray region corresponds to the 95% confidence interval. Each day of the year was reconstructed with MIMGP. The red dashed line represents the linear interpolation considering all the dates tagged "clear," and the black dotted line represents the smoothed signal using the Whittaker smoother.

temporal acquisition (the blue point at the top of Fig. 11). A clear drop of the reflectance can be seen in the gap-filled infrared (red dashed line). The reconstructed reflectance for MIMGP (black line) and Whittaker smoother (black dotted line) does not exhibit such a drop in the reflectance. Similar comments can be done for the second pixel in Fig. 11 (bottom), which has an undetected saturation (pixel value equal to zero). Other wavelengths are reported in the Supplementary Material.

VIII. CONCLUSION

This work introduces a novel approach to jointly classify and reconstruct irregularly sampled multidimensional time series. The proposed model, namely MIMGP, only involves a small number of parameters and is scalable to large datasets. The performances of the method were illustrated on a full year of Sentinel-2 satellite image time-series dataset from 2018 involving a high number of temporal acquisitions. A by-product of MIMGP is also to infer a confidence interval on the reconstruction.

The reconstruction has shown a good behavior on noisy pixels, while the model does not compete with the state-of-the-art classifiers, such as RF or SVM. One limitation comes from the independence hypothesis between spectral wavelengths, which is not true in practice. Therefore, our future work will be dedicated to the definition of multivariate GPs able to consider the correlation between wavelengths.

Another work-in-progress concerns the estimation of the mean function. MIMGP is able to handle the irregular sampling caused by multiple tiles, but the mean function is assumed to be spatially stationary. Rather than using a linear combination of fixed basis functions, a more complex approximating function depending on the geographical location could be used.

Finally, this work can be extended in a number of directions that could be of interest in the remote sensing field. A first extension would be the use of the MIMGP model to perform classification in the unsupervised case due to an EM-like algorithm. A second direction of future research would be the joint use of both Sentinel-2 and Sentinel-1 time series due to the MIMGP model, which allows for arbitrary temporal sampling in each spectral band.

APPENDIX A
TIME-SERIES RECONSTRUCTION

Let us first consider the case where the class of the missing value is known to be c. In such a situation, the reconstruction can be achieved using the conditional expectation

$$\hat{Y}^j_{b,c}(t^*) := \mathbb{E}\Big[Y^j_b(t^*)\big|\mathbf{y}_j, Z_j = c\Big].$$

The independence assumption A2 then yields

$$\mathbb{E}\Big[Y^j_b(t^*)\big|\mathbf{y}_j, Z_j = c\Big] = \mathbb{E}\Big[Y^j_b(t^*)\big|\mathbf{y}_{j,b}, Z_j = c\Big].$$

Besides, conditionally on $Z_i = c$, one has

$$\begin{pmatrix} Y^{\ j}_b(t^*) \\ \mathbf{y}_{j,b} \end{pmatrix} \sim \mathcal{N}_{T_j+1}\left(\begin{bmatrix} \mathbf{b}^*\boldsymbol{\alpha}_{b,c} \\ \mathbf{B}^j\boldsymbol{\alpha}_{b,c} \end{bmatrix}, \begin{bmatrix} K(t^*, t^*|\boldsymbol{\theta}_{b,c}) & \mathbf{k}\left(t^*, t^j_{1:T_j}|\boldsymbol{\theta}_{b,c}\right)^\top \\ \mathbf{k}\left(t^*, t^j_{1:T_j}|\boldsymbol{\theta}_{b,c}\right) & \boldsymbol{\Sigma}^j(\boldsymbol{\theta}_{b,c}) \end{bmatrix}\right)$$

where $\mathbf{k}(t^*, t_{1:T_j}|\boldsymbol{\theta}_{b,c}) = [K(t^*, t_1|\boldsymbol{\theta}_{b,c}), \ldots, K(t^*, t_{T_j}|\boldsymbol{\theta}_{b,c})]^\top$, $\mathbf{b}^* = [\varphi_1(t^*), \ldots, \varphi_J(t^*)]$, and $\mathbf{B}^j$ is the $T_j \times J$ design matrix associated with the time series $\mathbf{y}_j$ such that $(\mathbf{B}^j)_{\ell,k} = \varphi_k(t^j_\ell)$

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for all $(\ell, k) \in \{1, \ldots, T_j\} \times \{1, \ldots, J\}$. From classical properties of conditional Gaussian distributions (see [53, p. 63]), it follows that:

$$\mathbb{E}\Big(Y_b^j(t^*)\big|\mathbf{y}_{j,b};\, Z_j = c\Big) = \mathbf{b}^*\boldsymbol{\alpha}_{b,c} + \mathbf{k}\Big(t^*, t^j_{1:T_j}\big|\boldsymbol{\theta}_{b,c}\Big)\boldsymbol{\Sigma}^j(\boldsymbol{\theta}_{b,c})^{-1}\big(\mathbf{y}_{i,b} - \mathbf{B}^j\boldsymbol{\alpha}_{b,c}\big)$$

$$\mathbb{V}\Big(Y_b^j(t^*)\big|\mathbf{y}_{j,b};\, Z_j = c\Big) = K\big(t^*, t^*\big|\boldsymbol{\theta}_{b,c}\big) - \mathbf{k}\Big(\boldsymbol{\theta}_{b,c}\big|t^*, t^j_{1:T_j}\Big)\boldsymbol{\Sigma}^j(\boldsymbol{\theta}_{b,c})^{-1}\mathbf{k}\Big(t^*, t^j_{1:T_j}\big|t\boldsymbol{\theta}_{b,c}\Big)^\top.$$

Finally, replacing the unknown quantities by their estimated counterparts yields the desired results.

Second, when the class is unknown, the reconstruction rule is given by

$$\hat{Y}_b^j(t^*) = \mathbb{E}\Big[Y_b^j(t^*)\big|\mathbf{y}_j\Big].$$

Since

$$\mathbb{E}\Big[Y_b^j(t^*)\big|\mathbf{y}_j\Big] = \sum_{c=1}^{\mathcal{C}} \mathbb{P}\big(Z_j = c\big|\mathbf{y}_j\big)\mathbb{E}\Big[Y_b^j(t^*)\big|\mathbf{y}_j, Z_j = c\Big]$$

it straightforwardly follows that:

$$\hat{Y}_b^j(t^*) = \sum_{c=1}^{\mathcal{C}} \mathbb{P}\big(Z_j = c\big|\mathbf{y}_j\big)\hat{Y}_{b,c}^j(t^*)$$

where the reconstructed value $\hat{Y}_{b,c}^j(t^*)$ is provided in (11), while the posterior probabilities are given by (10). The variance is obtained by a similar calculation

$$\begin{aligned}\mathbb{V}\Big(Y_b^j(t^*)\Big) &= \mathbb{E}\Big[Y_b^j(t^*)^2\big|\mathbf{y}_j\Big] - \mathbb{E}\Big[Y_b^j(t^*)\big|\mathbf{y}_j\Big]^2 \\ &= \sum_{c=1}^{\mathcal{C}} \mathbb{P}\big(Z_j = c\big|y_j\big)\mathbb{E}\Big[Y_b^j(t^*)^2\big|\mathbf{y}_j, Z_j = c\Big] - \hat{Y}_b^j(t^*)^2.\end{aligned}$$

Besides, remarking that

$$\begin{aligned}\mathbb{E}\Big[Y_b^j(t^*)\big|\mathbf{y}_j, Z_j = c\Big] &= \hat{Y}_{b,c}^j(t^*) \\ \mathbb{E}\Big[Y_b^j(t^*)^2\big|\mathbf{y}_j, Z_j = c\Big] &= \mathbb{V}\Big(Y_{b,c}^j(t^*)\big|\mathbf{y}_j, Z_j = c\Big) + \mathbb{E}\Big[Y_{b,c}^j(t^*)\big|\mathbf{y}_j, Z_j = c\Big]^2\end{aligned}$$

it follows that:

$$\mathbb{V}\Big(\hat{Y}_b^j(t^*)\Big) = \sum_{c=1}^{\mathcal{C}} \mathbb{P}\big(Z_j = c\big|y_j\big)\mathbb{V}\Big(\hat{Y}_{b,c}^j(t^*)\big|\mathbf{y}_j, Z_j = c\Big) + \sum_{c=1}^{\mathcal{C}} \mathbb{P}\big(Z_j = c\big|y_j\big)\hat{Y}_{b,c}^j(t^*)^2 - \hat{Y}_b^j(t^*)^2$$

and the result is proved. Let us highlight that these derivations were conducted when a single time t^* is considered. Similar calculations can be achieved when reconstructed simultaneously several values. In such a case, this estimation procedure provides the covariance matrix of the reconstructed values at each time.

ACKNOWLEDGMENT

The authors would like to thank the reviewers for their many constructive and helpful comments. They would like to thank S. Iovleff for his support and advice during the construction of the model. They would also like to thank Y. Tanguy for his help when using the CNES computational resources to run the experiments presented in this article.

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